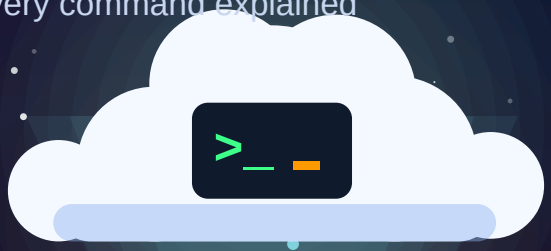
AWS | LINUX | BEGINNER GUIDE

# LAUNCH & USE A LINUX SERVER ON AWS EC2

A step-by-step guide with every command explained



Amazon Linux 2023

EC2 Instance Connect

```
$ sudo su
# yum update -y
  Complete!
# mkdir mayur
# touch my
```



INSTANCE TYPE

**t3.micro**

2 vCPU | 1 GiB memory

Free Tier eligible

| 10 | 23 | 15 MIN |
|---|---|---|
| REAL SCREENSHOTS | COMMANDS EXPLAINED | BEGINNER FRIENDLY |

AMAZON LINUX 2023 | T3.MICRO | EC2 INSTANCE CONNECT

# Overview

This guide walks you through creating a virtual Linux server in the cloud using **Amazon EC2**, connecting to it from your web browser, and running your first Linux commands. It follows the exact sequence shown in your screenshots and explains what every setting and command does.

## What you will accomplish

- Launch an **Amazon Linux 2023** server (EC2 instance) named **My-linux-connect**.
- Understand each launch setting: AMI, instance type, key pair, network, firewall and storage.
- Connect to the server from the browser using **EC2 Instance Connect** (no software needed).
- Run six essential commands: **`sudo su`**, **`yum update -y`**, **`mkdir`**, **`ls`**, **`cd`** and **`touch`**.

## Contents

| Part | Topic | What it covers |
|---|---|---|
| **1** | Launch the instance | Console, name, AMI, instance type, key pair, network, storage, launch |
| **2** | Connect to the instance | Connection methods, EC2 Instance Connect, username, public IP |
| **3** | Run Linux commands | Every command from your session, explained line by line |
| **4** | Command cheat sheet | Extra everyday commands to practise next |
| **5** | Security & cleanup | Hardening tips and how to stop charges |
| **6** | Troubleshooting | Common problems and fixes |

## Before you start

- An **AWS account** with a payment method (the Free Tier covers t3.micro for eligible accounts).
- A modern web browser. No SSH client or key file is needed for this method.
- About 15 minutes.

> **TIP**
> Your screenshots use the **Europe (Stockholm)** region. Always check the region selector at the top-right of the AWS console, because instances live in one region and you will only see them there.

# Part 1 - Launch the EC2 Instance

An **EC2 instance** is a virtual computer running in AWS data centres. You rent it by the hour and can create or delete it at any time.

## 1 Open the EC2 service and click Launch instance

Sign in to the AWS Management Console, search for **EC2**, and open it. On the EC2 home page, click the orange **Launch instance** button in the *Launch a virtual server* card.

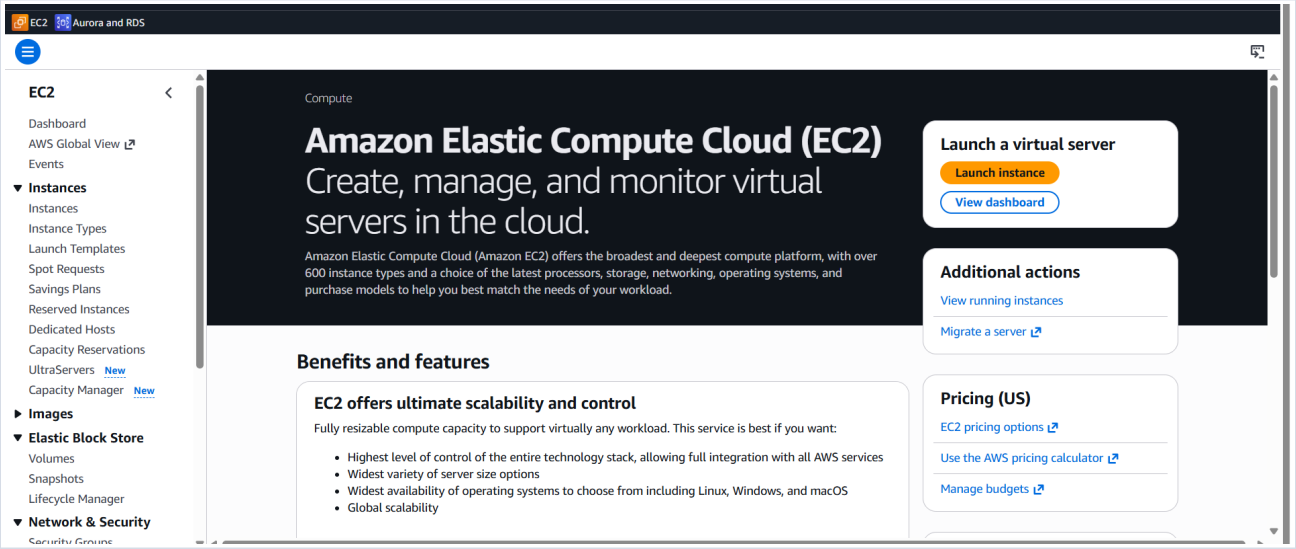



*Figure 1 - The EC2 home page. The Launch instance button is on the right.*

## 2 Name your instance

Under **Name and tags**, type a name. Here it is **My-linux-connect**. The name is stored as a tag (Name = My-linux-connect) and helps you find the server later in a list of many.

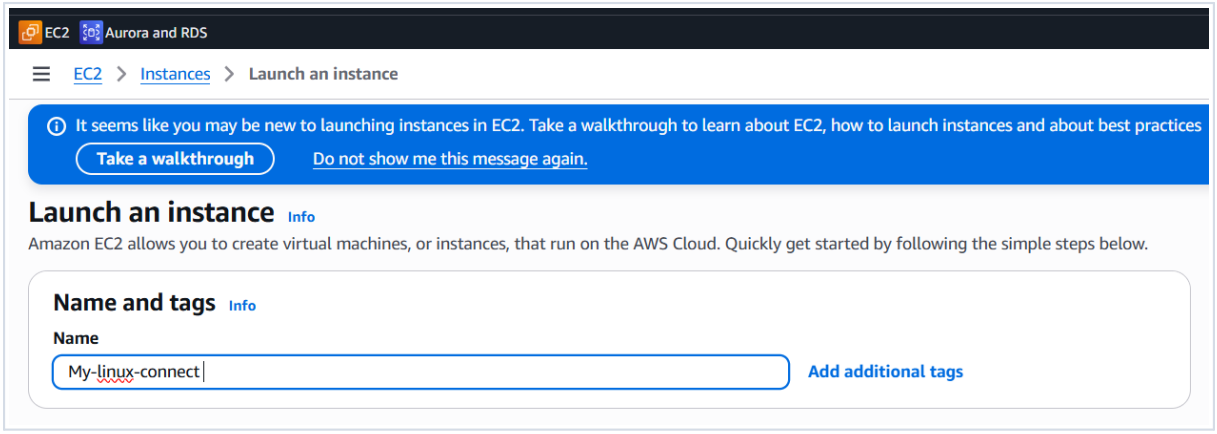



*Figure 2 - Naming the instance in the Name and tags section.*

## 3 Choose the operating system (AMI)

An **Amazon Machine Image (AMI)** is a template that contains the operating system and pre-installed software. On the **Quick Start** tab, **Amazon Linux** is selected, and the AMI is **Amazon Linux 2023 (kernel 6.18)**, marked *Free tier eligible*.

- **Architecture: 64-bit (x86)** - the standard Intel/AMD processor type; keep this to match t3.micro.
- **Username: ec2-user** - the default login name for this AMI. You will need it when connecting.
- **Verified provider** - confirms the image is published by Amazon.

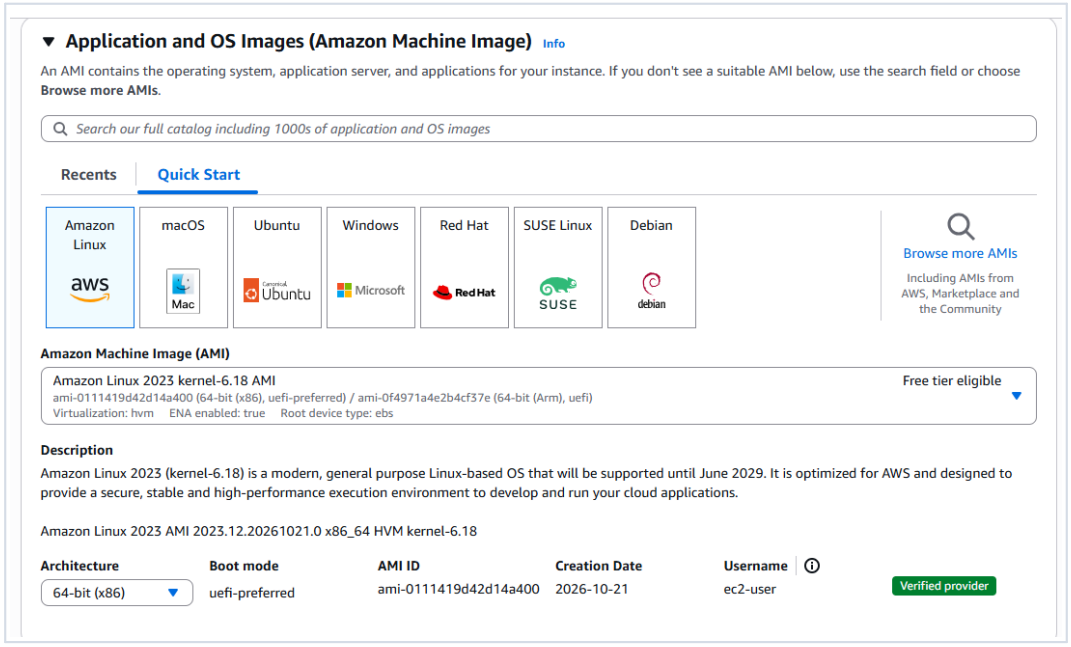



*Figure 3 - Selecting Amazon Linux 2023 from Quick Start.*

## 4 Choose the instance type

The **instance type** sets the CPU and memory. Select **t3.micro** (2 vCPU, 1 GiB memory), which is Free Tier eligible and plenty for learning. On-demand Linux pricing is shown at about 0.0108 USD per hour, so always stop or terminate instances you are not using.

## 5 Key pair (login)

A **key pair** is a private file used to log in over SSH. In the screenshot, *Proceed without a key pair* is chosen, which is fine here because you connect through **EC2 Instance Connect** in the browser, which creates temporary keys automatically. If you plan to use PuTTY or a terminal later, select **Create new key pair** and keep the downloaded file safe.

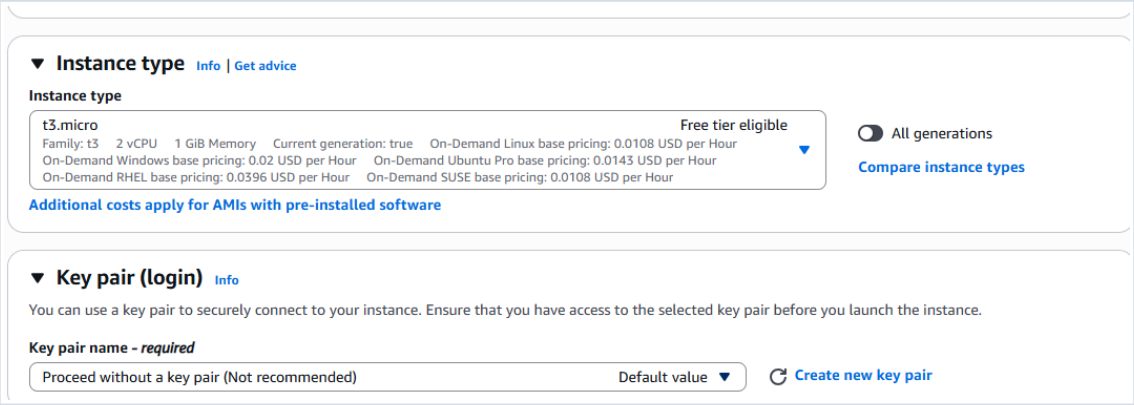



*Figure 4 - Instance type t3.micro and the key pair setting.*

## 6 Network settings and firewall

Leave the defaults and review the firewall. A **security group** is a virtual firewall for your instance. The wizard creates one named **launch-wizard-1** with these rules:

- **Allow SSH traffic (port 22)** - required for EC2 Instance Connect and SSH logins.
- **Allow HTTPS traffic (port 443)** and **HTTP traffic (port 80)** - only needed if you will host a website.
- **Auto-assign public IP: Enable** - gives the server an internet-reachable address.

> **SECURITY WARNING**
> The source **Anywhere (0.0.0.0/0)** lets every IP address on the internet attempt to reach port 22. For anything beyond a short test, change it to **My IP** so only your own address can connect. Uncheck HTTP/HTTPS if you are not running a web server.

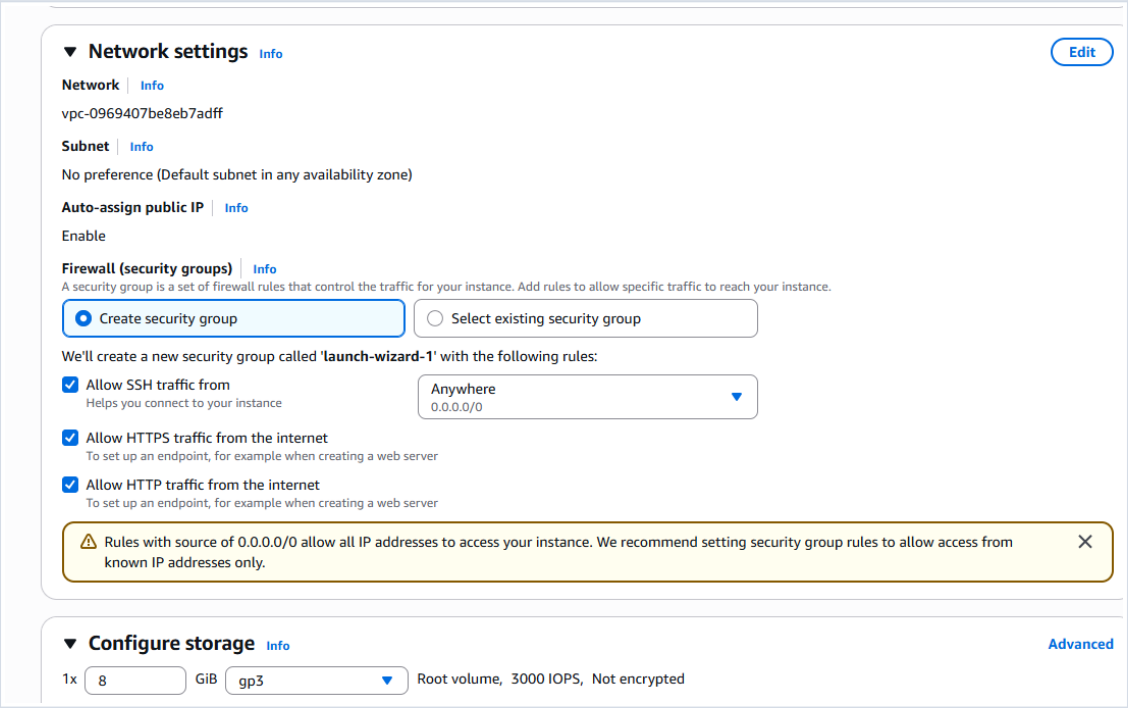



*Figure 5 - Network settings, security group rules, and the default 8 GiB gp3 storage.*

## 7 Configure storage

The default is a single **8 GiB gp3** root volume (3000 IOPS, not encrypted). That is the server's hard disk and is enough for this exercise. Under **File systems** keep **None** selected, since you are not attaching shared storage such as EFS or FSx.

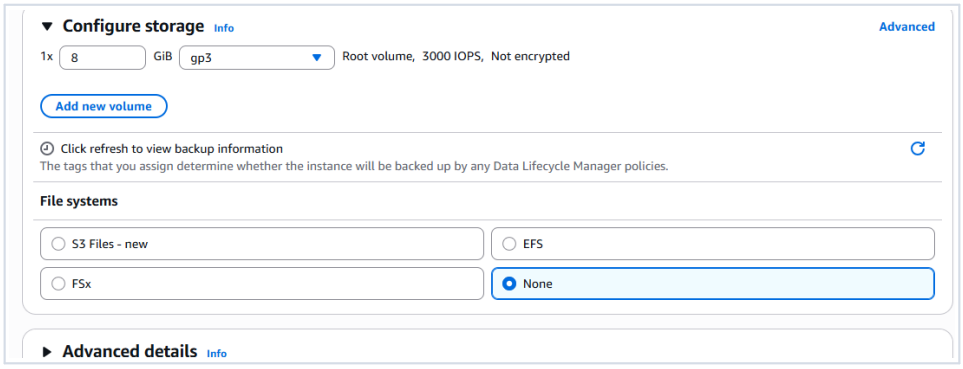



*Figure 6 - Storage and file system options.*

## 8 Review the summary and launch

The **Summary** panel on the right confirms your choices: 1 instance, Amazon Linux 2023 AMI, t3.micro, a new security group and 1 volume of 8 GiB. When everything looks right, click **Launch instance**.

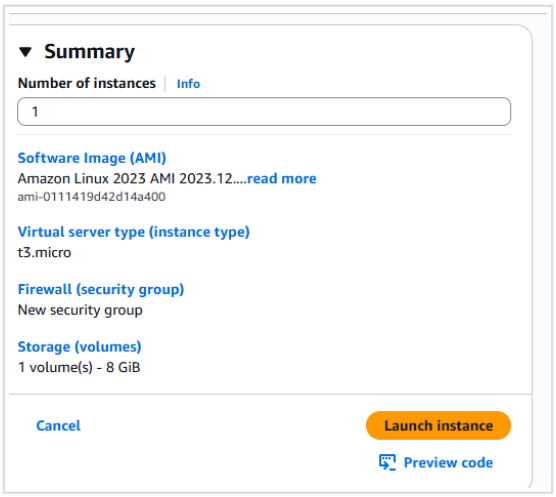



*Figure 7 - Summary panel with the Launch instance button.*

## 9 Confirm success

A green **Success** banner appears with your new **instance ID** (a unique identifier beginning with *i-*). The *Next Steps* area offers optional extras such as billing alerts and monitoring. For now, click **Connect to instance**.

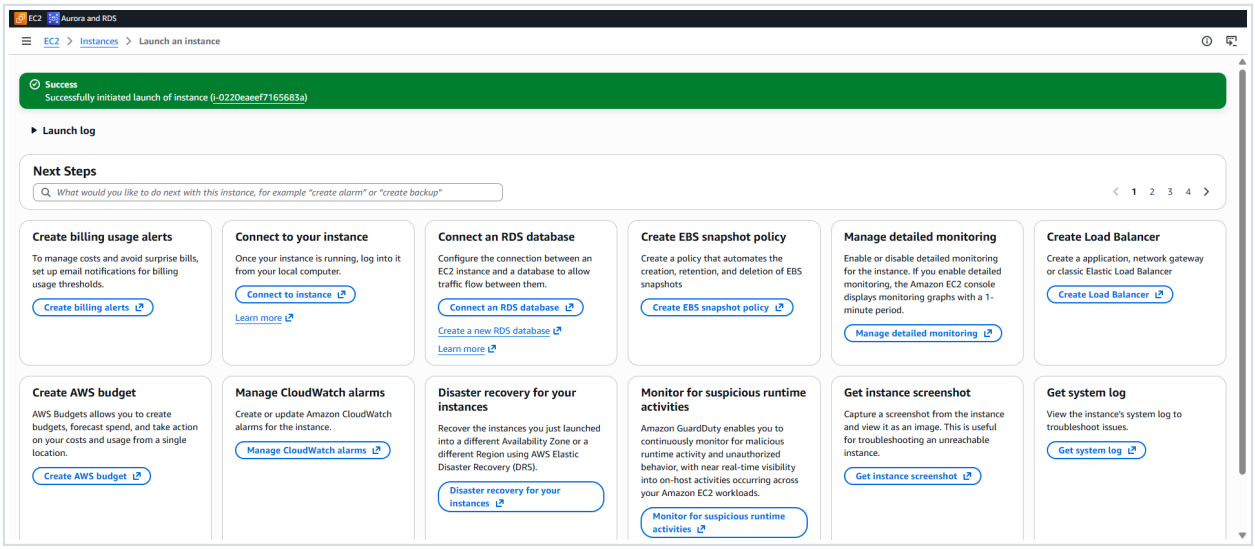



*Figure 8 - Launch confirmation and Next Steps.*

**RECOMMENDED**

Use **Create billing alerts** or **Create AWS budget** to be notified about unexpected charges.

# Part 2 - Connect to the Instance

The **Connect to Linux instance** page offers four ways to log in. This guide uses the first one.

| Method | Best for | Needs |
|---|---|---|
| **EC2 Instance Connect** (used here) | Quick browser terminal for instances with a public IP | Port 22 open; no key file |
| EC2 Instance Connect Endpoint | Instances in private subnets | A VPC endpoint |
| SSM Session Manager | Highest security; no open ports | IAM role and SSM agent |
| EC2 Serial Console | Fixing boot or network problems | Account-level enablement |

## 10 Connect with EC2 Instance Connect

- Select the **In web browser** tab and keep **EC2 Instance Connect** selected.
- Under **Choose public IP address**, keep the **IPv4 address** option.
- Confirm the **Username** is **ec2-user**.
- Click the orange **Connect** button. A new browser tab opens with a terminal.

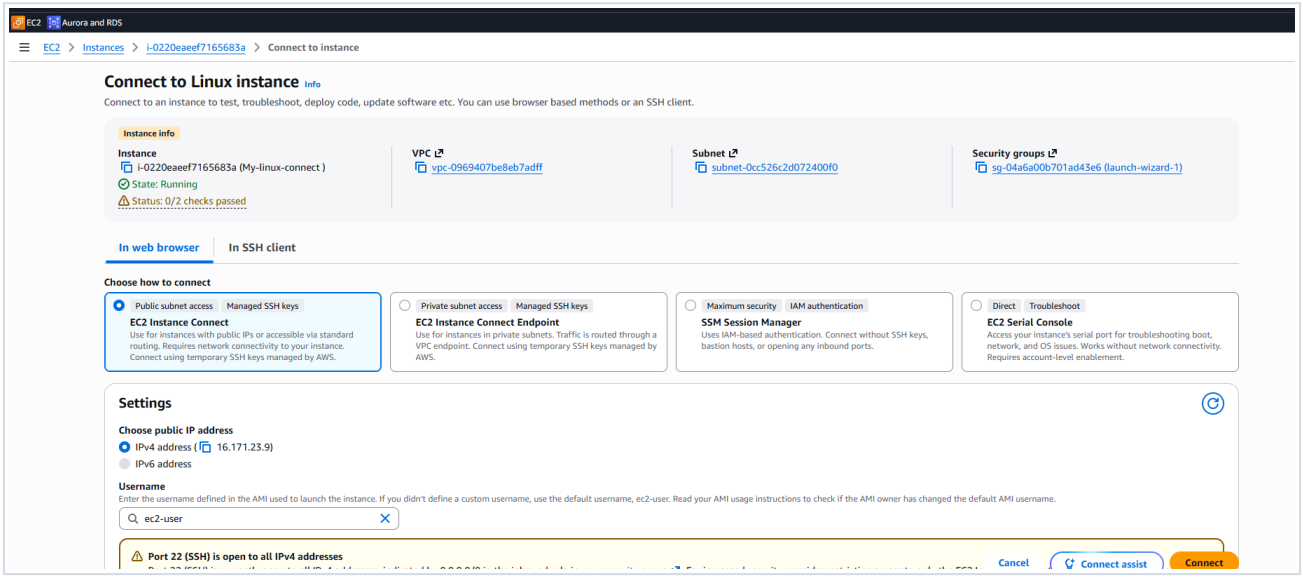



*Figure 9 - Connect to Linux instance page (EC2 Instance Connect).*

> **NOTE**
> Right after launch, the status may show **0/2 checks passed** while AWS finishes initialising. Wait one to two minutes, refresh, and connect once the checks pass.

# Part 3 - Run Linux Commands (Explained)

Once connected you see the Amazon Linux 2023 banner and a **prompt**. Everything you type after the prompt is a command; press **Enter** to run it.

## Reading the prompt

```
[ec2-user@ip-172-31-39-125 ~]$
```

| Part | Meaning |
|---|---|
| `ec2-user` | The user you are logged in as. |
| `ip-172-31-39-125` | The server's host name (derived from its private IP address). |
| `~` | Your current folder. **~** is shorthand for your home directory. |
| **$** / # | **$** means a normal user; # means the administrator (root). |

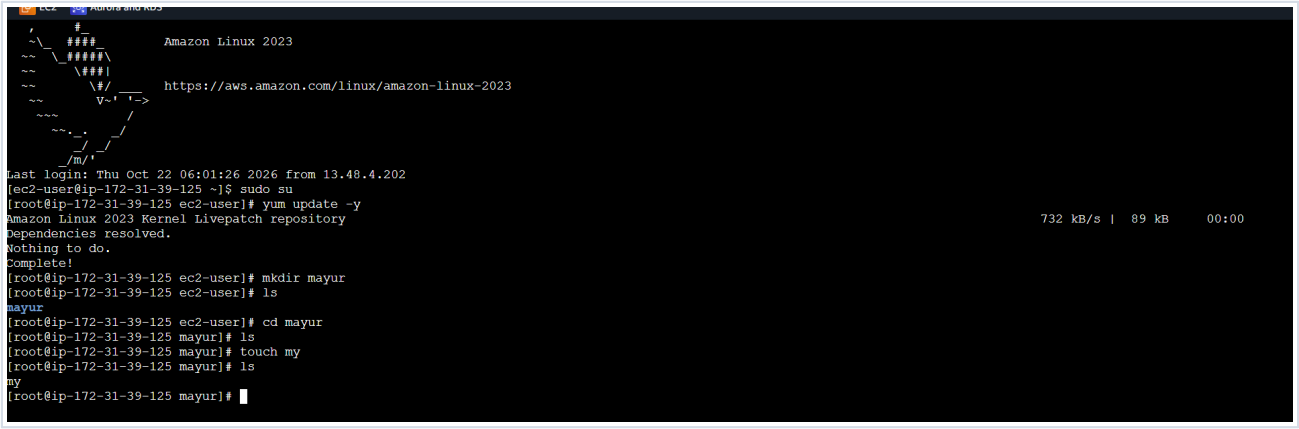



*Figure 10 - Your terminal session (account header removed).*

## The commands, one by one

### 1 `sudo su`

**What it does:** Switches you to the **root** (administrator) account.

- **sudo** = "superuser do": runs a command with administrator rights.
- **su** = "switch user": with no name given, it switches to root.
- The prompt changes from **[ec2-user@... ~]$** to **[root@... ec2-user]#**. The # confirms you are root.
- **Why use it:** installing and updating software needs root rights.
- **Caution:** root can change or delete anything. Prefer running single commands as **`sudo command`** in real work.

```
[ec2-user@ip-172-31-39-125 ~]$ sudo su
[root@ip-172-31-39-125 ec2-user]#
```

## 2 `yum update -y`

**What it does:** Checks the software repositories and updates every installed package to its latest version.

- **yum** is the package manager (it installs, updates and removes software). On Amazon Linux 2023 it is an alias for **dnf**, so both work.
- **update** tells it to upgrade installed packages.
- **-y** answers "yes" automatically so it does not pause to ask for confirmation.
- In your screenshot the result was **Nothing to do. Complete!**, meaning the fresh server was already up to date.

```
# yum update -y
Dependencies resolved.
Nothing to do.
Complete!
```

## 3 `mkdir mayur`

**What it does:** Creates a new directory (folder) named **mayur** in the current location.

- **mkdir** = "make directory".
- The argument **mayur** is the folder name. Linux names are **case-sensitive**, so *Mayur* and *mayur* are different.
- No output means success; Linux stays quiet when a command works.

```
# mkdir mayur
```

## 4 `ls`

**What it does:** Lists the files and folders in the current directory.

- **ls** = "list".
- After **`mkdir`**, it shows **mayur** (blue text marks a folder).
- Run again inside the empty folder, it prints nothing, because the folder has no contents yet.
- Useful options: **`ls -l`** (detailed list), **`ls -a`** (include hidden files), **`ls -la`** (both).

```
# ls
mayur
```

## 5 `cd mayur`

**What it does:** Moves you into the **mayur** directory.

- **cd** = "change directory".
- The prompt now ends with **mayur]#** showing your new location.
- Handy variants: **`cd ..`** (up one level), **`cd ~`** (home), **`cd /`** (top of the file system).

```
[root@ip-172-31-39-125 ec2-user]# cd mayur
[root@ip-172-31-39-125 mayur]#
```

## 6 `touch my`

**What it does:** Creates a new, empty file named **my**.

- **touch** normally updates a file's timestamp, but if the file does not exist it creates an empty one.
- Running **`ls`** afterwards shows **my**, confirming the file now exists inside **mayur**.
- You can create several at once: **`touch a.txt b.txt c.txt`**.

```
# touch my
# ls
my
```

## Full session at a glance

```
$ sudo su                  # become root
# yum update -y            # update all software
# mkdir mayur              # create a folder
# ls                       # list  ->  mayur
# cd mayur                 # enter the folder
# ls                       # list  ->  (empty)
# touch my                 # create an empty file
# ls                       # list  ->  my
```

# Part 4 - Command Cheat Sheet

Practise these next. They build directly on the commands you just used. Commands marked **caution** change or delete data.

| Command | Meaning | Example |
|---|---|---|
| **pwd** | Print working directory: shows where you are | `pwd` |
| **whoami** | Shows the user you are logged in as | `whoami` |
| **ls -l** | Long list with permissions, owner, size, date | `ls -l` |
| **cd ..** | Go up one folder | `cd ..` |
| **cat** | Display a file's contents | `cat notes.txt` |
| **echo** | Print text; with > it writes text into a file | `echo hello > my` |
| **cp** | Copy a file | `cp my my-backup` |
| **mv** | Move or rename a file | `mv my new-name` |
| **rm (caution)** | Delete a file. There is no recycle bin | `rm my-backup` |
| **rm -r (caution)** | Delete a folder and everything in it | `rm -r mayur` |
| **nano** | Simple text editor in the terminal | `nano notes.txt` |
| **df -h** | Show free disk space | `df -h` |
| **free -h** | Show memory usage | `free -h` |
| **clear** | Clear the screen | `clear` |
| **history** | Show commands you have run | `history` |
| **man** | Open the manual page for a command | `man ls` |
| **exit** | Leave root, then close the session | `exit` |

> **TIP**
> Press the **Tab** key to auto-complete file names and the **Up arrow** to bring back previous commands. Use **`command --help`** for quick help on almost any command.

# Part 5 - Security and Cleanup

## Security best practices

- **Restrict SSH** to **My IP** instead of 0.0.0.0/0 in the security group.
- **Avoid working as root.** Use **sudo** for individual commands only.
- **Keep software updated** with **`sudo yum update -y`**.
- **Encrypt volumes** for anything sensitive (the default root volume here is not encrypted).
- **Protect key files** if you create a key pair, and never share them.

### Stop charges when you finish

| Action | Effect | Billing |
|---|---|---|
| **Stop** instance | Shuts the server down; the disk and data are kept | No compute charge; the 8 GiB disk still costs a little |
| **Terminate** instance | Permanently deletes the server and its root disk | All charges stop |

To do either, open **EC2 > Instances**, select **My-linux-connect**, choose **Instance state**, then **Stop instance** or **Terminate instance**.

> **IMPORTANT**
> Terminating is permanent. Copy anything you need off the server first.

# Part 6 - Troubleshooting

| Problem | Likely cause | Fix |
|---|---|---|
| Connect button fails or times out | Port 22 is blocked or the instance is still starting | Check the security group allows SSH, and wait until status checks show 2/2 passed. |
| Permission denied | Wrong username | Use **ec2-user** for Amazon Linux (Ubuntu uses **ubuntu**). |
| *command not found* | Typo or software not installed | Check spelling and case; install with **`sudo yum install name`**. |
| *Permission denied* when editing system files | Not running as root | Prefix the command with **sudo**. |
| Cannot find the instance | Looking in the wrong region | Switch the region selector (e.g. Europe (Stockholm)). |

> **CONGRATULATIONS**
> You have launched a cloud server, logged in, and created your first folder and file. A good next step is installing a web server: **`sudo yum install httpd -y`**.